

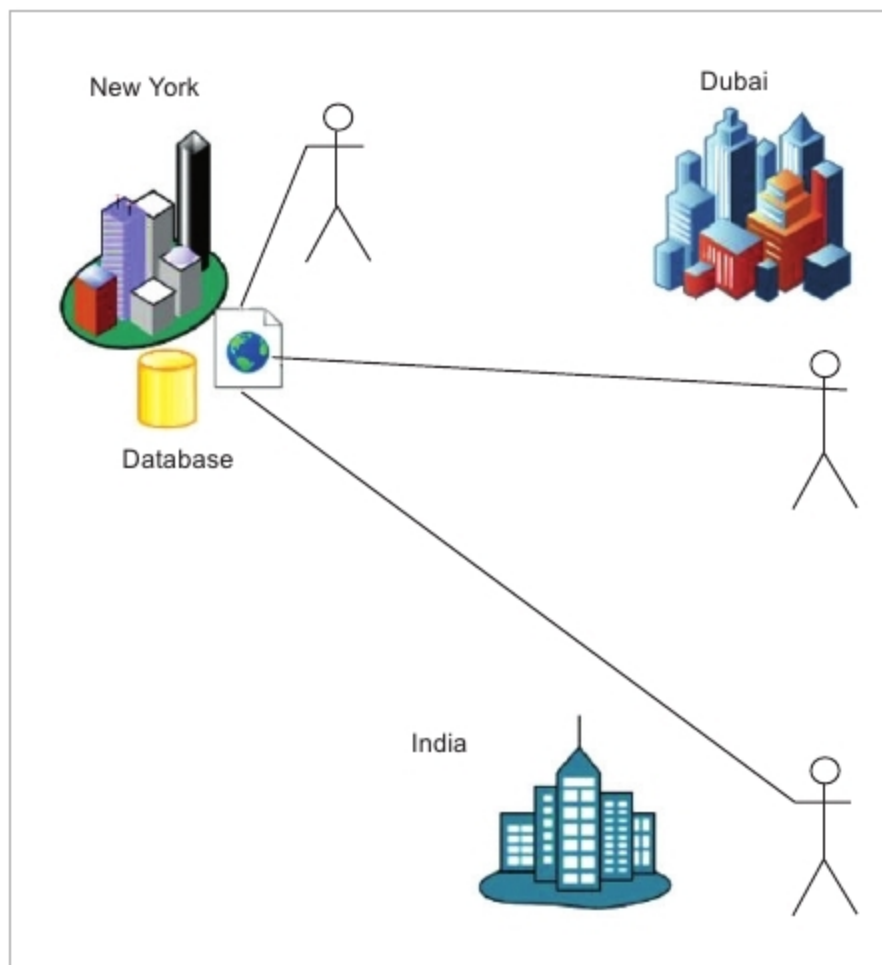


Figure 1: Managing data without Cosmos DB

- It is elastic, globally scalable and with HA, automatically indexes all data.
- It has five consistency levels — bounded staleness, consistent prefix, session consistency, eventual consistency and immediate consistency. And there are more options to choose between consistency and performance.

Figure 1 shows a simple example of geographical data that highlights the importance of and need for Cosmos DB.

Managing data without Cosmos DB:

- Data geo-replication might be a challenge for the developer.
- Users from remote locations might experience latency and inconsistency in their data.
- Providing an automatic failover is a real challenge.

Managing data with Cosmos DB:

- Data can be geo-distributed with just a few clicks.
- Developers do not need to worry about data replication.
- Strong consistency can be given to the end users across geo-distributed locations.
- Web-tier application can be changed anytime between primary and secondary tiers, just with a few clicks.
- Failover can be initiated any time manually, and automatic failover is present.

The five consistency levels

Data consistency is configurable on Cosmos DB, letting application developers choose from five different levels:

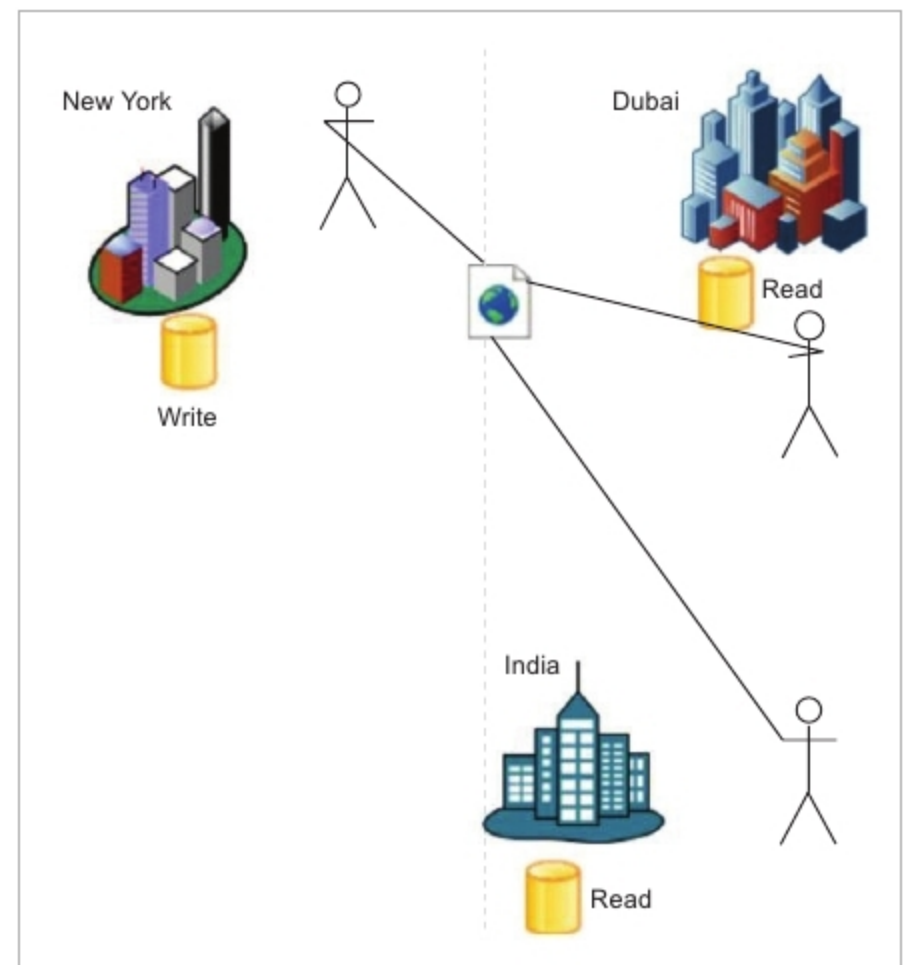


Figure 2: Managing data with Cosmos DB

- *Eventual consistency* doesn't guarantee any ordering and only ensures that replicas will eventually converge.
- *Consistent prefix* adds ordering guarantees on top of eventual consistency.
- *Session consistency* is scoped to a single client connection, and basically ensures a read-your-own-writes consistency for each client; it is the default consistency level.
- *Bounded staleness* augments consistent prefix by ensuring that reads won't lag beyond x versions of an item or some specified time window.
- *Strong consistency* (or linearisability) ensures that clients always read the latest globally committed write.

Until now, Microsoft has been a follower on the NoSQL DB path, with MongoDB as the leader. However, with the new release of Cosmos DB and the strong community evolving around developing applications for this database, it is fast acquiring a share in the NoSQL DB market. The use of data migration tools also makes the import of data from other databases to Cosmos DB rather easy. So Cosmos DB is here to stay and is a very good offering from Microsoft. Let's see what the tech giant offers next. **END** 🐧

Reference

https://en.wikipedia.org/wiki/Cosmos_DB

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